# Risk-Aware Active Perception for Crowd Navigation with a Pan Camera: From Replayed Crowds to Photorealistic RGB-D Simulation

Author One, Author Two, Author Three
Affiliation
email@domain

***Abstract*—A robot with a narrow-field-of-view camera cannot track every pedestrian around it, so it must decide where to look while it moves. We couple gaze control with motion planning in a single chance-constrained model predictive controller. The camera is pointed to shrink the uncertainty of the collision constraints that matter for the planned path, including pedestrians not yet detected by a Poisson multi-Bernoulli tracker. On 1,200 paired episodes of replayed ETH/UCY crowds, active gaze beats a fixed camera and look-at-nearest on Zara (with smaller, non-significant gains on ETH/Hotel) and performs on par with information-gain gaze. In a pre-registered SocNavBench study with rendered RGB-D images, a person detector and data association, the method succeeds in 70% of episodes against 53% for a fixed camera ($p_{\text{Holm}} = 0.005$) and 56% for information-gain gaze ($p_{\text{Holm}} = 0.06$).**

## I. Introduction

Crowd navigation planners usually assume that all nearby pedestrians are observed. A robot with an onboard camera sees only a cone in front of it: people approaching from the side are detected late, and tracks of people who leave the view become stale. Active perception, choosing where to look, is classically driven by information gain [1], which reduces uncertainty regardless of whether it matters for the task. We ask whether gaze driven by the *collision risk of the planned motion* navigates crowds more safely.

## II. Method

**Planner.** We extend SupeR-MPC [2], a robust MPC with chance constraints that accounts for state-estimation, parametric and process uncertainty. The state is the unicycle pose plus a camera pan angle $\phi$, and the inputs are $(v, \omega, \omega_\phi)$. The horizon is $N{=}8$ steps of 0.4 s.

**Tracking and prediction.** A Poisson multi-Bernoulli (PMB) tracker [3] keeps Bernoulli tracks with existence probability $r_j$, plus a Poisson intensity of undetected pedestrians. The intensity is advected by flow channels learned from the scene, and tracks whose position standard deviation exceeds 1 m are pruned. A Social LSTM [4], trained leave-one-scene-out with rotation augmentation, predicts Gaussian trajectories; the propagated track covariance is added to the network's.

**Chance constraints.** For tracked pedestrian $j$ at step $k$,

$$a_{jk}^\top (c_{r,k} - \mu_{jk}) - r_{\text{c}} \geq \kappa_j \sqrt{2\, a_{jk}^\top (\Sigma_{r,k} + \Sigma_{jk})\, a_{jk}}, \qquad (1)$$

with $\kappa_j = \operatorname{erf}^{-1}(1 - 2\min(\delta/r_j, \tfrac{1}{2}))$, so unlikely tracks are constrained less. Undetected pedestrians get one chance constraint per step on the expected undetected mass inside the robot's collision disk. It is *observation-aware*: a mass counts only if the planned camera views before the robot arrives would not have detected it, which couples speed to what the camera can see. Constraints always use current information, so safety never relies on views that have not been taken yet.

**Risk-aware gaze.** The cost rewards views that will shrink future constraint back-offs. Let $\text{vis}_{jk}$ be a smooth visibility model and $\pi_{jm}$ the probability that pedestrian $j$ is first seen at step $m$. The expected covariance is then

$$\begin{aligned}\tilde{\Sigma}_{jk} = \Sigma_{jk}^{\text{model}} &+ \textstyle\prod_{m\leq k}(1 - \text{vis}_{jm})\, \Sigma_{jk}^{\text{state}} \\ &+ \textstyle\sum_{m\leq k} \pi_{jm} J_{k-m} P_\infty J_{k-m}^\top,\end{aligned} \qquad (2)$$

and the cost adds $w\, \rho_{jk} \operatorname{erf}^{-1}(1-2\delta) \sqrt{2 a_{jk}^\top (\Sigma_{r,k} + \tilde{\Sigma}_{jk}) a_{jk}}$ for the most collision-relevant pedestrians ($\rho_{jk}$: relevance to the warm-start plan). A second term penalizes the undetected mass left in the path's collision disks after the planned views. Weights and directions are fixed from the warm start, so only gaze can lower the cost.

## III. Experiments

**Protocol.** Success means reaching the goal (0.3 m) within the time budget without collision. Paired methods run on identical episodes and are compared with exact McNemar tests and Holm correction. Confirmation hypotheses were fixed before running, on unused seeds. Baselines differ only in gaze: fixed, look-at-nearest, and information gain ($\log\det$ of expected track covariances).

**Replayed crowds.** The robot crosses ETH/UCY scenes (60° FOV, 4 m range) among replayed, non-reactive pedestrians. The camera model includes occlusion, a 0.9 detection probability and range-dependent noise. The PMB prior is fitted on the first half of each recording, and episodes are drawn from the second half (600 per dataset group). On Zara,

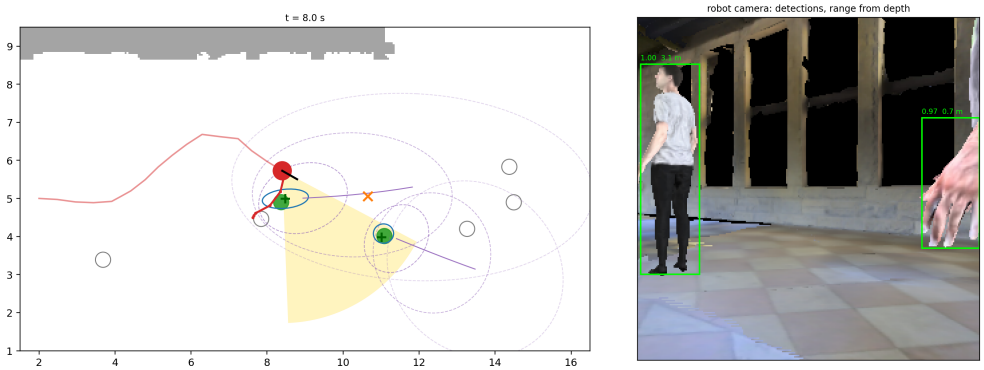



Fig. 1. SocNavBench Zara episode with RGB-D perception. Left: true pedestrians (green: detected), PMB tracks (blue), predictions (purple), the camera cone and the robot's plan. Right: the rendered robot view with detections (score, depth range).

TABLE I
SUCCESS ON REPLAYED CROWDS (600 PAIRED EPISODES PER COLUMN). ORACLES: GROUND-TRUTH GAZE; 360° CAMERA.

| Gaze | Zara01/02 | ETH/Hotel |
|---|---|---|
| Risk-aware (ours) | **0.632** | 0.610 |
| Information gain | 0.637 | **0.630** |
| Look at nearest | 0.555 | 0.610 |
| Fixed | 0.547 | 0.570 |
| Gaze oracle | 0.660 | 0.677 |
| 360° camera | 0.852 | 0.797 |

active gaze beats look-at-nearest ($p_{\text{Holm}}$=0.005) and the fixed camera ($p$<0.001), with half as many collisions with never-seen pedestrians (125 vs. 223 and 212; Table I). On ETH/Hotel it ties look-at-nearest and leads the fixed camera without significance ($p$=0.08). Information-gain gaze performs on par in both scene groups ($p$=0.89 and 0.39). An ablation (run with an earlier predictor) shows that the PMB tracker, active gaze and stale-track pruning each significantly raise success.

**Photorealistic RGB-D simulation.** We connected the planner to SocNavBench [5]. It replays the same recordings in S3DIS building scans [6], with pedestrians rendered as SURREAL bodies [7]. Each step, the robot renders a 256×256 RGB-D image from its pose and pan angle (Fig. 1). A COCO-trained Faster R-CNN [8] detects persons, the range comes from the depth pixels inside each box, and detections are associated with the tracks by gated Hungarian matching. The robot pose is still ground truth plus noise. On rendered frames, detection precision is 0.93 and recall is 0.88–0.96 within 4 m, with a median position error of 0.12 m. Walls enter the MPC as half-plane constraints at the nearest map points. We use SocNavBench's 30 curated single-recording episodes with three new seeds each (90 paired episode–seed runs). Collisions with pedestrians who appeared less than 2 s earlier are excluded: the recordings start tracks mid-scene, sometimes on top of the robot.

With real vision, risk-aware gaze beats the fixed camera (won:lost 19:4, $p_{\text{Holm}}$=0.005; per episode, averaged over seeds: 15:2). It also leads information-gain gaze by 14 points (24:11), but that comparison misses significance after correction ($p_{\text{Holm}}$=0.061, Table II). Unlike in the replays, where

TABLE II
SOCNAVBENCH CONFIRMATION RUN (30 EPISODES × 3 SEEDS). SUCCESS EXCLUDING SPAWN COLLISIONS; $p$: ONE-SIDED EXACT MCNEMAR VS. OURS, HOLM-CORRECTED.

| Gaze | Sim. camera | RGB-D vision | $p_{\text{Holm}}$ (vision) |
|---|---|---|---|
| Risk-aware (ours) | **0.70** | **0.70** | – |
| Information gain | 0.58 | 0.56 | 0.061 |
| Fixed | 0.56 | 0.53 | 0.005 |

every route crosses the main flow, the lead over information gain also appears with the simulated camera: SocNavBench routes mix crossing, following and opposing traffic around obstacles, so fewer pedestrians matter at once and relevance-weighted gaze pays off (a hypothesis we have not yet tested). Real vision lowered no method's success, so perception noise was not the bottleneck at this range.

## IV. CONCLUSION

Coupling gaze to the collision risk of the planned motion beats passive sensing on replayed Zara crowds and, at a pre-registered corrected level, with real RGB-D perception in a photorealistic simulator. Its advantage over information-gain gaze depends on the scenario: none when every pedestrian crosses the robot's path, and a consistent but not yet significant lead in mixed traffic around obstacles. Limitations, and next steps, are non-reactive replayed pedestrians, ground-truth localization and the short detection range.

## REFERENCES

[1] R. Bajcsy, "Active perception," *Proc. IEEE*, vol. 76, no. 8, pp. 966–1005, 1988.
[2] Zhu, Simeon, and Cognetti, "SupeR-MPC," *IEEE Robot. Autom. Lett.*, 2025.
[3] J. L. Williams, "Marginal multi-Bernoulli filters: RFS derivation of MHT, JIPDA, and association-based MeMBer," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 51, no. 3, pp. 1664–1687, 2015.
[4] A. Alahi *et al.*, "Social LSTM: Human trajectory prediction in crowded spaces," in *Proc. CVPR*, 2016.
[5] A. Biswas *et al.*, "SocNavBench: A grounded simulation testing framework for evaluating social navigation," *ACM Trans. Human-Robot Interact.*, vol. 11, no. 3, 2022.
[6] I. Armeni *et al.*, "3D semantic parsing of large-scale indoor spaces," in *Proc. CVPR*, 2016.
[7] G. Varol *et al.*, "Learning from synthetic humans," in *Proc. CVPR*, 2017.
[8] S. Ren *et al.*, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. NeurIPS*, 2015.